

File Formats

4.1 Raw

A raw file is the unprocessed data from your camera's sensor. Very few point-and-shoot cameras offer the raw format as an output option. One such example is Panasonic's Lumix DMC-LX3. Many dSLRs offer the option of raw output, but in different formats depending on the make and model of the camera. There are a wide range of extensions for raw files, which are often undocumented so that only the camera can work on the data. The raw file, is simply put, the image as captured by the sensor, without artificial corrections like interpretation of the picture from the sensor (demosaijing), white balance, brightness and contrast. These aspects of the image are processed after the image is captured on the sensor.

For many professionals, the raw file is analogous to the negative in digital photography, and the wide proliferation of formats is a big problem. For example, Canon uses a CRW extension, Sony uses SRF and Panasonic uses a RAW extension. Most algorithms used to store and compress the raw data are proprietary, and consumers do not have access to them.

Saving files in the raw format allows you to capture everything possible from the camera, and then correct the image or process it later in the camera itself or using an image editing software. This means that the raw data has much more information about the image, and much more subtle gradations in the saturation and hue of the image, than in the compressed formats. Also, the raw format lacks any of the artefacts inherent in image compression to the JPEG and TIFF formats. However, raw data takes up more memory space than standard compressed files.

Raw files should be used primarily if you expect extensive editing work for the photo you are clicking. This means that changing say the brightness of the image, or the saturation of particular colours, will come out with a much higher dynamic range than it would have with other formats.